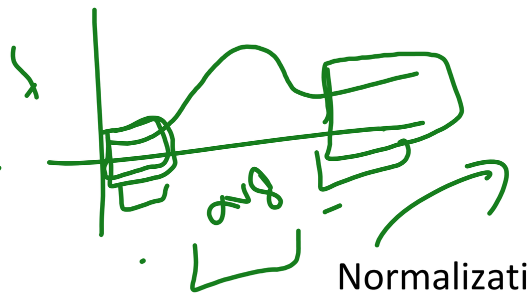
90%



min, max

Scaing

$\boxed{0 \quad 255}$

When to use

Standardization

Use when:

- Data is roughly **Gaussian (normal-like)**
- You care about **distance / gradients / optimization**

Common cases

- Neural Networks
- Logistic Regression
- Linear Regression
- SVM
- PCA

Why

- Handles **outliers better than normalization**
- Makes gradients stable

I

Normalization

Use when:

- You need values in a **fixed range (0–1)** 0,1
- No assumption about distribution
- Features have **known bounds**

Common cases

- Image data (pixel values 0–255 → 0–1)
- KNN
- K-means
- When using distance-based models with bounded data

Why

- Preserves shape but **rescales extremes**
- Useful when scale must be controlled

Important difference (intuition)

Standardization:

```
keeps outliers → just rescales
distribution
```

Normalization:

```
compresses everything into range →
outliers squash others
```